

Agenda

1. Friday Feedback
2. Trig Substitution
3. Summary of
Integration Techniques



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Recall:

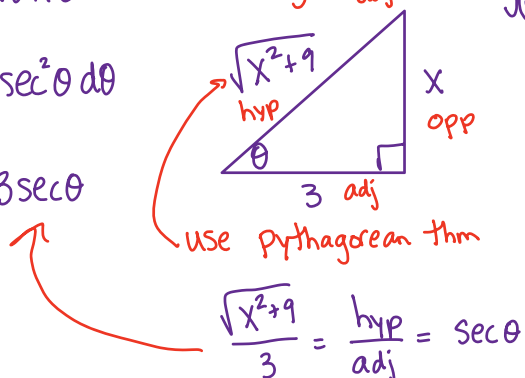
Integrand contains	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$ $a^2 - (a \sin \theta)^2 = (a \cos \theta)^2$
$\sqrt{x^2 + a^2}$	$x = a \tan \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$\tan^2 \theta + 1 = \sec^2 \theta$ $(a \tan \theta)^2 + a^2 = (a \sec \theta)^2$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, 0 \leq \theta < \frac{\pi}{2}$ OR $\pi < \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$ $(a \sec \theta)^2 - a^2 = (a \tan \theta)^2$

Activity: $\int \frac{dx}{(x^2+9)^{3/2}} = \int \frac{dx}{(\sqrt{x^2+9})^3}$ $\sqrt{x^2+a^2}$ with $a=3$

$x = 3 \tan \theta$ $\tan \theta = \frac{x}{3} = \frac{\text{opp}}{\text{adj}}$

$dx = 3 \sec^2 \theta d\theta$

$\sqrt{x^2+9} = 3 \sec \theta$



$\int \frac{dx}{(\sqrt{x^2+9})^3} = \int \frac{3 \sec^2 \theta}{(3 \sec \theta)^3} d\theta$

$= \int \frac{3 \sec^2 \theta}{27 \sec^3 \theta} d\theta$

$= \frac{1}{9} \int \frac{1}{\sec \theta} d\theta$

$= \frac{1}{9} \int \cos \theta d\theta$

$= \frac{1}{9} \sin \theta + C = \frac{x}{9\sqrt{x^2+9}} + C$

Example: $\int (4x^2+9)^{1/2} dx = \int \sqrt{(2x)^2+9} dx = 2 \int \sqrt{x^2+\frac{9}{4}}$

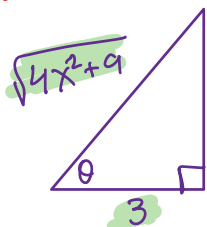
$2x = 3 \tan \theta \rightarrow \frac{2x}{3} = \tan \theta$

$x = \frac{3}{2} \tan \theta$

$dx = \frac{3}{2} \sec^2 \theta d\theta$

$\sqrt{4x^2+9} = 3 \sec \theta$

$\sqrt{4x^2+9} = \sqrt{9 \tan^2 \theta + 9}$



$\frac{\sqrt{4x^2+9}}{3} = \frac{\text{hyp}}{\text{adj}} = \sec \theta$

$= \int (3 \sec \theta) \left(\frac{3}{2} \sec^2 \theta \right) d\theta$

$= \frac{9}{2} \int \sec^3 \theta d\theta$

(in book)

$= \frac{9}{4} \left[\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta| \right] + C$

$= \frac{9}{4} \left[\frac{\sqrt{4x^2+9}}{3} \cdot \frac{2x}{3} + \ln \left| \frac{\sqrt{4x^2+9}}{3} + \frac{2x}{3} \right| \right] + C$

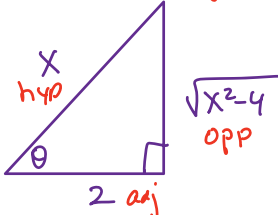
Activity: $\int \frac{\sqrt{x^2-4}}{x} dx$

$x = 2\sec\theta$

$\text{arcsec}\left(\frac{x}{2}\right) = \theta$
 $\sec\theta = \frac{x}{2} = \frac{\text{hyp}}{\text{adj}}$

$dx = 2\sec\theta\tan\theta d\theta$

$\sqrt{x^2-4} = 2\tan\theta$



$\frac{\sqrt{x^2-4}}{2} = \frac{\text{opp}}{\text{adj}} = \tan\theta$

$$\int \frac{2\tan\theta}{2\sec\theta} \cdot 2\sec\theta\tan\theta d\theta$$

$$= 2 \int \tan^2\theta d\theta$$

$$= 2 \int (\sec^2\theta - 1) d\theta$$

$$= 2\tan\theta - 2\theta + C$$

$$= \sqrt{x^2-4} - 2\text{arcsec}\left(\frac{x}{2}\right) + C$$

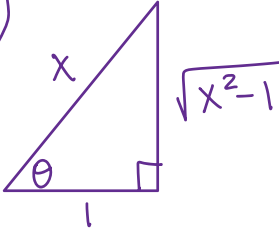
Example: $\int_{x=1}^{x=3} \frac{dx}{\sqrt{x^2-1}}$

$x = \sec\theta \rightarrow \frac{x}{1} = \sec\theta$

$dx = \sec\theta\tan\theta d\theta$

$\sqrt{x^2-1} = \tan\theta$

$\theta = \text{arcsec}(x)$



$\frac{\sqrt{x^2-1}}{1} = \frac{\text{opp}}{\text{adj}} = \tan\theta$

$$= \int_{\text{arcsec}(1)}^{\text{arcsec}(3)} \frac{\sec\theta\tan\theta}{\tan\theta} d\theta$$

$$= \int_{\text{arcsec}(1)}^{\text{arcsec}(3)} \sec\theta d\theta$$

formula sheet

$$= \ln|\sec\theta + \tan\theta| \Big|_{\text{arcsec}(1)}^{\text{arcsec}(3)}$$

$$= \ln|x + \sqrt{x^2-1}| \Big|_1^3$$

$$= \ln(3+\sqrt{8}) - \ln(1+\sqrt{0})$$

$$= \ln(3+\sqrt{8})$$

Review of Integration Techniques

Integration by Parts: $\int u dv = uv - \int v du$
 $\int_a^b u dv = [uv]_a^b - \int_a^b v du$

- * Choose dv so that it is easily integrated
Choose u so that $\int v du$ is simpler than $\int u dv$
- * sometimes we have to do IBP again (again)
 $\int x^3 \sin x dx$ $\int x^{100} e^{5x} dx$, $\int e^{2x} \sin(4x) dx$

Trig Integrals:

- powers of sine/cosine
- powers of secant/tangent
- $\sin(mx)\cos(nx)$

- * typical approach: use trig identities & u-sub
- * when in doubt turn everything in sine & cosine
 $\int \cos^4 \theta \tan \theta d\theta = \int \cos^4 \theta \cdot \frac{\sin \theta}{\cos \theta} d\theta \dots$

Trig Substitution: $\sqrt{a^2 - x^2}$, $\sqrt{x^2 + a^2}$, $\sqrt{x^2 - a^2}$
 $x = a \sin \theta$, $x = a \tan \theta$, $x = a \sec \theta$

* Use the Right triangle!

* answers should be algebraic
(not $\cot(\operatorname{arsec}(x))$)

Partial Fraction Decomposition Next week!

Activity: For each integral, identify the best technique
(basic, u-sub, IBP, trig int, trig sub, no tool yet)

$$\int \frac{\ln(x)}{x} dx \quad \int \tan^3 \theta \sec^4 \theta d\theta \quad \int \sin^3 \theta d\theta$$

$$\int \frac{1}{\sqrt{x^2 - 6x + 45}} \quad \int \frac{x^2 - 2x + 3}{x^2 - 1} dx \quad \int \cos^4 \theta d\theta$$

$$\int_0^1 \frac{t-4}{t^2-5t+6} \quad \int x^3 \sqrt{9-x^2} dx \quad \int (\ln(t))^2 dt$$

Example: $\int \frac{1}{\sqrt{x^2 - 6x + 45}} dx$